Week 5 Lab: Digital Implementation, Filtering, and Robustness

ENGR 3303 – Controls Engineering
Fall 2026

> **Main goal.** Improve the motor-speed controller by testing digital sample time, adding an RPM filter, and evaluating final controller robustness using Python data analysis.

Hardware and Software

Use the same hardware from Weeks 1–4:

- Arduino Uno or compatible board;
- L298N motor-driver module;
- 12 V encoder gearmotor;
- external motor power supply;
- Arduino IDE;
- Python with pyserial, pandas, NumPy, and Matplotlib.

Install Python packages:

```
pip install pyserial pandas numpy matplotlib scipy
```

> **Safety**
> - Keep the wheel clear of fingers, wires, and clothing.
> - Do not touch the spinning wheel directly.
> - Stop testing if the motor becomes rough, oscillatory, or repeatedly saturates.
> - Do not apply disturbances unless approved by the instructor.

Course encoder calibration: Use `COUNTS_PER_OUTPUT_REV = 494.0` for Encoder A counted with `RISING` edges only. Recalibrate only if the motor or encoder-counting method changes.

Lab 5A – Sample Time and Filtered RPM

Suggested time: 75 minutes

Part 1 – Controller Settings

Use your selected PI gains from Week 4.

$$K_p = _____, \qquad K_i = _____$$

Use the same reference speed as before:

$$\boxed{r = _____ \text{ RPM}}$$

Part 2 – Upload the Filtered Controller Code

Open:

`week5_pi_filtered_controller.ino`

Before uploading, verify:

- `REFERENCE_RPM`;
- `KP`;
- `KI`;
- `SAMPLE_MS`;
- `FILTER_ALPHA`;
- `COUNTS_PER_OUTPUT_REV`.

The Arduino sends:

time, reference, raw RPM, filtered RPM, error, PWM, integral

Part 3 – Baseline Test with No Filtering

Use:

$$\alpha = 1.0.$$

This means no filtering.
Run:

`week5_record_filtered_control.py`

Save the file as:

`filter_alpha1.csv`.

Part 4 – Test Filtered RPM

Now choose a filtered case, such as

$$\alpha = 0.5$$

or

$$\alpha = 0.7.$$

Save the data as:

`filter_alpha0p5.csv`

or another clear filename.

> **Lab 5A checkoff:** Show the instructor a plot comparing raw RPM and filtered RPM.

Part 5 – Compare Filtering

Complete:

α	Final RPM	Error	Max PWM	Observation
1.0				

Lab 5B – Robustness and Final Controller Test

Suggested time: 75 minutes

Part 1 – Choose Final Controller Settings

Choose one sample time and one filter value.

$$\boxed{T_s = _____ \text{ s}, \qquad \alpha = _____}$$

Also record:

$$\boxed{K_p = _____, \quad K_i = _____, \quad r = _____ \text{ RPM}}$$

Part 2 – Run Three Repeated Trials

Run the same final controller three times.

Save:

`final_trial1.csv`

`final_trial2.csv`

`final_trial3.csv`

Run:

`week5_compare_final_trials.py`

This script compares repeated trials and creates a summary table.

Part 3 – Optional Disturbance Test

With instructor approval only, apply a small safe disturbance.
Record:

- when the disturbance occurred;
- how much the RPM dropped;
- how quickly the controller recovered;
- whether PWM saturated.

Do not touch the spinning wheel directly.

Part 4 – Final Controller Summary

Complete the table.

Trial	Final RPM	Error	Overshoot	Max PWM	Observation
1					
2					
3					

Questions to Submit

Answer briefly.

1. What sample time did your final controller use?
2. What filter value α did your final controller use?
3. How did filtering change the measured RPM signal?
4. Did filtering make the controller response slower?
5. What final K_p, K_i values did your team use?
6. What was the average steady-state error over three trials?
7. Did the controller behave consistently across trials?
8. Did PWM saturate? If yes, when?
9. If you performed a disturbance test, how did the motor recover?
10. What is one thing you would improve if you had more time?

What to Submit

One submission per team:

- CSV data for filtering tests;
- CSV data for three final-controller trials;
- raw-vs-filtered RPM plot;
- repeated-trial comparison plot;
- completed tables;
- answers to Questions 1–10;
- Arduino and Python code.

End-of-Lab Shutdown

Before leaving:

1. set PWM to zero;
2. turn OFF motor power;
3. save all CSV, Arduino, and Python files;
4. disconnect USB after motor power is off;
5. return hardware in an organized condition.